



Computational Chemistry Comparison and Benchmark DataBase

Release 22 (May

2022) Standard Reference Database 101 [National Institute of Standards and Technology](https://cccbdb.nist.gov/exp2x.asp)

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Experimental data for NO (Nitric oxide)

22 02 02 11 45

Other names			
Amidogen, oxo-; Nitric oxide; Nitrogen monoxide; Nitrogen oxide; Nitrogen oxide (NO); Nitrosyl radical;			
INChI	INChIKey	SMILES	IUPAC name
InChI=1S/NO/c1-2	MWUXSHHQAYIFBG-UHFFFAOYSA-N	[N]=O	Nitrogen monoxide
State	Conformation		
$^2\Pi$	C*V		

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H^\circ$	91.04	0.08	kJ mol ⁻¹	2006Rus/Pin:6592	
Hfg(0K) $\Delta_f H^\circ$	90.54	0.08	kJ mol ⁻¹	2006Rus/Pin:6592	
Entropy (298.15K) S°	210.76		J K ⁻¹ mol ⁻¹	JANAF	
Integrated Heat Capacity (0 to 298.15K) $H^\circ - H^\circ(T)$	9.19		kJ mol ⁻¹	JANAF	
Heat Capacity (298.15K) C_p°	29.86		J K ⁻¹ mol ⁻¹	webbook	

Information can also be found for this species in the [NIST Chemistry Webbook](#)

Vibrational levels (cm⁻¹)

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm ⁻¹)	Harmonic(cm ⁻¹)	Reference	(km mol ⁻¹)	unc.	Reference		
1	Σ	1876	1904	2007Iri:389					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
1904.135	14.08836	0.0100467	1.704888	1.754158E-02	948.6467	2007Iri:389

vibrational zero-point energy: 938.0 cm⁻¹ (from fundamental vibrations)[Calculated vibrational frequencies](#) for NO (Nitric oxide).

More spectroscopic constants are available at the NIST Physics Laboratory website:

<http://physics.nist.gov/PhysRefData/MolSpec/Diatomic/Html/Tables/NO.html>

Rotational Constants (cm⁻¹)

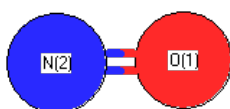
See section [L.F.4 to change rotational constant units](#)

A	B	C	reference	comment
	1.67195		1979HUB/HER	

[Calculated rotational constants](#) for NO (Nitric oxide).

Product of moments of inertia $I_A I_B I_C$		
10.08262	amu Å ²	1.674284E-39 gm cm ²

Geometric Data



Point Group $C_{\infty v}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rNO	1.154	0.000	1	2			NISTdiatomic	r0

Cartesians

Atom	x (Å)	y (Å)	z (Å)
O1	0.0000	0.0000	0.0000
N2	0.0000	0.0000	1.1508

Atom - Atom Distances

Distances in Å

	O1	N2
O1		1.1508
N2	1.1508	

[Calculated geometries](#) for NO (Nitric oxide).

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
N=O	1

Connectivity

Atom 1	Atom 2
O1	N2

Electronic energy levels (cm⁻¹)

Energy (cm ⁻¹)	Degeneracy	reference	description
0	2	1979HUB/HER	² Π
119.8	2	1979HUB/HER	
38440	8	1979HUB/HER	
43965.7	2	1979HUB/HER	

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
9.264	0.000	9.264		webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
0.026	0.005	webbook

Proton Affinity (kJ mol⁻¹)

Proton Affinity	unc.	Product	reference	comment
531.8		HNO ⁺	webbook	
		NOH ⁺		

Dipole, Quadrupole and Polarizability**Electric dipole moment $\leftrightarrow$**

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	² Π	C _{∞v}	True				0.159	1971Byf/Car:271	MB μ ₀	C _{∞v}	1	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

[Calculated electric dipole moments](#) for NO (Nitric oxide).

Electric quadrupole moment $+^-+$

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	² Π	C _{∞v}	True						C _{∞v}	1	1

[Calculated electric quadrupole moments](#) for NO (Nitric oxide).

Electric dipole polarizability (Å³) $\leftrightarrow$

alpha	unc.	Reference
1.698		1997Oln/Can:59

[Calculated electric dipole polarizability](#) for NO (Nitric oxide).

References

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squib	reference	DOI
1971Byf/Car:271	CR Byfleet, A Carrington, DK Russell, "Electric dipole moments of open-shell diatomic molecules" Molecular Physics 1971, 20(2), 271-277	10.1080/00268977100100251
1979HUB/HER	Huber, K.P.; Herzberg, G., Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules, Van Nostrand Reinhold Co., 1979	10.1007/978-1-4757-0961-2
1997Oln/Can:59	TN Olney, NM Cann, G Cooper, CE Brion, Absolute scale determination for photoabsorption spectra and the calculation of molecular properties using dipole sum-rules, Chem. Phys. 223 (1997) 59-98	10.1016/S0301-0104(97)00145-6
2006Rus/Pin:6592	B Ruscic, RE Pinzon, ML Morton, NK Srinivasan, M-C Su, JW Sutherland, JV Michael "Active Thermochemical Tables: Accurate Enthalpy of Formation of Hydroperoxyl Radical, HO ₂ " J. Phys. Chem. A 2006, 110, 6592-6601	10.1021/jp056311j
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
JANAF	Chase, M.W., Jr.; Davies, C.A.; Downey, J.R., Jr.; Frurip, D.J.; McDonald, R.A.; Syverud, A.N., JANAF Thermochemical Tables (Third Edition), J. Phys. Chem. Ref. Data, Suppl. 1, 1985, 14, 1.	
NISTdiatomic	NIST Diatomic Spectral Database (www.physics.nist.gov/PhysRefData/MolSpec/Diatomic/index.html)	10.18434/T4T59X
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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GeSe (Germanium monoselenide)	NO- (nitric oxide anion)